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## **EXPERIMENT 17**

### **PROLOG PROGRAM – SUM OF INTEGERS FROM 1 TO N**

#### **Aim**

To write and execute a **Prolog program to find the sum of integers from 1 to N** using recursion.

#### **Algorithm**

1. Start the program.
2. Read the value of N.
3. If  $N = 0$ , return the sum as 0.
4. Otherwise, calculate the sum of integers from 1 to  $N-1$ .
5. Add N to the obtained sum.
6. Continue the recursive process until N becomes 0.
7. Display the final sum.
8. Stop the program.

#### **Prolog Program**

% Sum of integers from 1 to N

sum\_to\_n(0, 0).

sum\_to\_n(N, Sum) :-

N > 0,

N1 is N - 1,

sum\_to\_n(N1, Sum1),

Sum is N + Sum1.

### Input

?- sum\_to\_n(10, Sum).

### Output:

Sum = 55 ? |

### Result

Thus, the **Prolog program to calculate the sum of integers from 1 to N was successfully written and executed**, and the correct sum was obtained.